

Student Employability & Placement Analytics Dashboard

Project Report

Prepared using the Campus Recruitment dataset (Kaggle, Ben Roshan)

215 students · 15 original columns

1. Introduction

Colleges routinely collect data on student academics, work experience, specialisation and placement outcomes, but this data is rarely converted into decisions anyone can act on. This project builds a data analytics dashboard that turns raw placement records into KPIs, trends, drivers, risks, opportunities and concrete recommendations for students and institutions.

2. Problem Statement

Given student-level academic, demographic and employability-related data, the project answers: What is the overall placement rate? What factors are associated with placement? How do academic performance, prior work experience and specialisation relate to outcomes? What drives package/salary differences among placed students? Which cohorts show weaker outcomes and need more support?

3. Objectives

- Calculate the overall placement rate and supporting KPIs
- Identify which academic and employability factors associate with placement
- Compare placement outcomes across degree streams and MBA specialisations
- Analyze salary/package patterns among placed students
- Translate findings into specific, data-backed recommendations

4. Dataset Description

Name: Campus Recruitment (Placement_Data_Full_Class.csv)

Source: Kaggle — "Campus Recruitment" dataset by Ben Roshan
(<https://www.kaggle.com/datasets/benroshan/factors-affecting-campus-placement>)

Size: 215 students, 15 columns (14 after dropping the row-index column)

Columns: gender; ssc_p / ssc_b (10th % / board); hsc_p / hsc_b / hsc_s (12th % / board / stream); degree_p / degree_t (degree % / stream); workex (prior work experience); etest_p (employability test %); specialisation (MBA specialisation); mba_p (MBA %); status (Placed / Not Placed); salary (package offered, placed students only).

Scope note: this dataset represents a single MBA cohort rather than multiple departments, and has no explicit certifications / extracurriculars field. Degree Stream and MBA Specialisation are used as the closest available department-equivalent groupings throughout this report and the dashboard; work experience and the employability test score serve as the closest available skill-development proxies.

5. Data Preprocessing

- Loaded the CSV and inspected shape (215 rows × 15 columns), dtypes and column names.

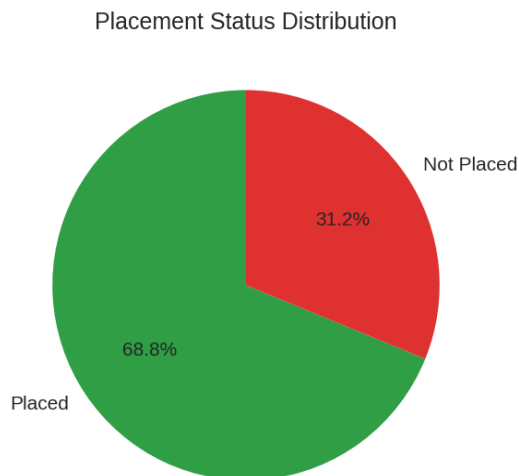
- Dropped the sl_no row-index column (no analytical value).
- Checked for duplicate records — none found.
- Checked for missing values — only `salary` has missing values (67 of 215 rows, 31.2%). This is structural, not a data-quality defect: students with status = "Not Placed" were never offered a salary. These rows are excluded from salary statistics rather than imputed with zero or a mean, which would distort the results.
- Checked categorical values for consistency (gender, board, stream, specialisation, workex, status) — all values were clean and consistently coded, no normalization needed.

6. Methodology

The analysis follows: **Data** → **KPIs** → **Trends** → **Drivers** → **Risks/Opportunities** → **Actionable Insights**. KPIs are computed directly from the (optionally filtered) dataset. Trends are identified via group comparisons (cross-tabs, group means). Drivers are identified using Pearson correlation between numeric academic/test scores and a binary placement flag, and by comparing placement rates across categorical segments (work experience, stream, specialisation). Findings are described as associations, not causal claims, since this is observational data.

7. Exploratory Data Analysis

7.1 Placement Status Distribution



148 of 215 students (68.8%) were placed; 67 (31.2%) were not.

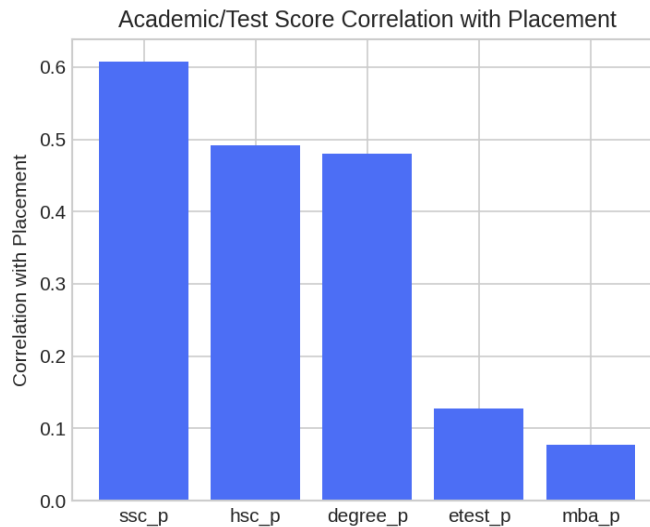
7.2 Academic Scores: Placed vs Not Placed

Status	ssc_p	hsc_p	degree_p	etest_p	mba_p
Not Placed	57.5	58.4	61.1	69.6	61.6

Placed	71.7	69.9	68.7	73.2	62.6
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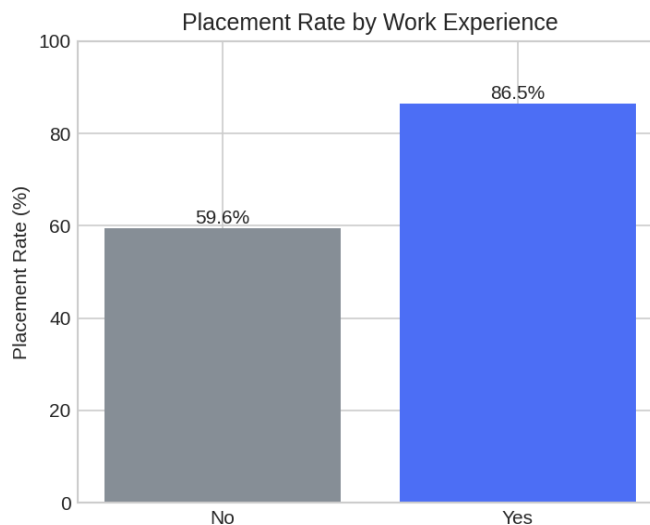
Placed students score higher on average at every academic stage tracked. The gap is widest at the 10th-grade stage (ssc_p) and narrows by the MBA stage (mba_p), suggesting early academic consistency is a stronger placement signal than late-stage performance in this cohort.

7.3 Correlation with Placement



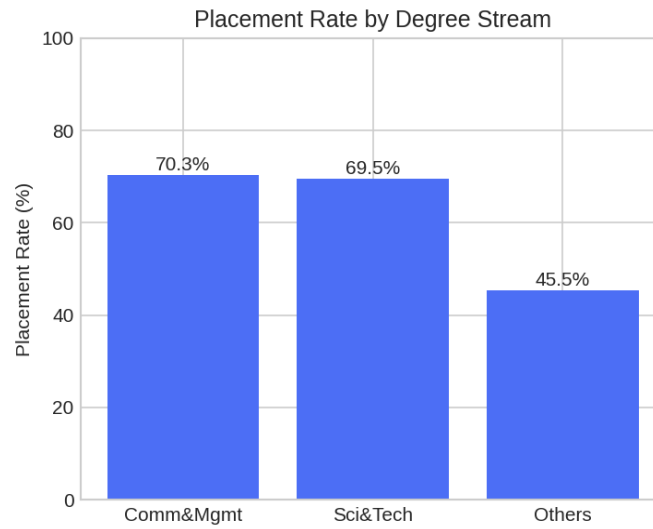
10th-grade percentage (ssc_p) shows the strongest correlation with placement ($r = 0.61$), followed by 12th-grade percentage (hsc_p, $r = 0.49$) and degree percentage (degree_p, $r = 0.48$). The MBA percentage and employability test score show much weaker correlation ($r = 0.08$ and $r = 0.13$ respectively).

7.4 Work Experience



Students with prior work experience are placed at 86.5%, compared to 59.6% for students without it — a gap of 26.9 percentage points, the largest single-factor gap found in this dataset.

7.5 Degree Stream



Comm&Mgmt; has the highest placement rate (70.3%), while Others trails at 45.5%.

8. KPI Summary

KPI	Value
Total Students	215
Placed Students	148
Placement Rate	68.8%
Average Degree %	66.4
Work Experience Participation	34.4%
Average Employability Test %	72.1
Average Package (Placed)	■288,655
Median Package (Placed)	■265,000

9. Trend Analysis

- Overall placement rate across the cohort is 68.8%.
- Comm&Mgmt; outperforms Others by 24.9 percentage points in placement rate.
- Median package (■265,000) is meaningfully lower than mean package (■288,655) — a small number of high outlier offers pull the average up.

10. Driver Analysis

- Prior work experience is the single strongest categorical driver of placement rate.
- 10th-grade percentage is the strongest numeric driver ($r = 0.61$), ahead of 12th-grade and degree percentage.
- MBA specialisation and degree stream both show meaningful placement-rate gaps between their best- and worst-performing categories.

11. Risks

- Students in the lowest-placing stream (Others) and specialisation are at higher risk of remaining unplaced.
- Package figures reported as a simple average can overstate typical student outcomes because of right-skew from a small number of high offers.
- Students without work experience face a materially lower placement rate and may need additional support to close that gap.

12. Opportunities

- Structured pre-placement internship programs are the highest-leverage lever available in this dataset, given the size of the work-experience placement gap.
- Mentoring or mock-interview pairing between Comm&Mgmt; and Others cohorts could help narrow the placement-rate gap.
- Publishing stream/specialisation-level placement rates lets institutions identify and support at-risk cohorts earlier in the year.

13. Actionable Insights

- Work experience before final placements is associated with the largest single improvement in placement likelihood in this dataset.
- Early academic consistency (10th/12th grade) matters more to placement outcomes here than a single late-stage test score — students should not rely on a strong final-year test to compensate for weaker early academics.
- Specialisation and stream choice correlate with materially different placement rates and should inform where institutions target extra support.

14. Recommendations

For Students

- Prioritise securing an internship or other work experience ahead of final placements.
- Build consistent academic performance from 10th grade onward rather than depending on a late-stage test score.
- If in a lower-placing specialisation or stream, seek targeted interview preparation and proactive employer outreach.

For Institutions

- Expand structured internship/work-experience pipelines, particularly for streams and specialisations with below-average placement rates.
- Track and publish stream-level placement rates to identify at-risk cohorts earlier.
- Pair strong- and weak-placement-rate cohorts for mentoring and mock-interview practice.
- Consider adding certification and extracurricular-activity tracking to future data collection — neither is captured in the current dataset, limiting analysis depth.

15. Machine Learning Model

No prediction model was built for this project. At 215 rows, a classifier would have limited statistical power and a high risk of overfitting to this specific cohort, and the project's goal is actionable, explainable insight rather than a black-box prediction. The correlation and group-comparison analysis in Sections 7 and 10

already identifies the same directional drivers a simple model would surface (work experience, early academic scores), without the added complexity and overfitting risk of a formal model on a dataset this small.

16. Technologies Used

- Python, Pandas, NumPy — data loading, cleaning and analysis
- Plotly — interactive charts in the dashboard
- Matplotlib — static charts for this report
- Streamlit — interactive web dashboard
- ReportLab — this PDF report

17. Dashboard Overview

The accompanying Streamlit dashboard (app.py) provides six pages: Overview (KPIs and headline charts), Academic Analysis, Employability Factors, Department/Stream Analysis, Salary Analysis, and Insights & Recommendations (auto-generated from whatever filters are currently applied). Sidebar filters for gender, degree stream, specialisation, placement status, work experience and degree-percentage range apply live across every page.

18. Results

The cohort's overall placement rate is 68.8%. Work experience is the single strongest categorical driver of placement outcome (86.5% vs 59.6% placement rate), and 10th-grade percentage is the strongest numeric driver ($r = 0.61$). Comm&Mgmt; and Mkt&Fin; specialisation post the strongest placement rates among their respective categories. Median package (₹265,000) is a more representative figure than mean package for typical student outcomes, given the right-skewed package distribution shown in the dashboard's Salary Analysis tab.

19. Conclusion

This project demonstrates that a modest, well-understood dataset can still yield concrete, actionable findings when analyzed with clear KPIs and driver analysis rather than description alone. Work experience and early academic consistency emerge as the clearest levers available to students and institutions in this cohort.

20. Future Scope

- Incorporate multi-year data to analyze placement trends over time rather than a single cohort snapshot.
- Add certification and extracurricular-activity fields to widen the analysis.

- Revisit a predictive model once the dataset grows well beyond 215 rows.

21. References / Dataset Source

Campus Recruitment dataset, Ben Roshan, Kaggle:

<https://www.kaggle.com/datasets/benroshan/factors-affecting-campus-placement>